VICTORIOUS JUBILATION

A Data Analytics and AI for Business Report

DECEMBER 15, 2022

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Introduction

Are you happy?

It's a deceptively simple question with potentially far-reaching implications. This question has plagued humanity's collective consciousness since the advent of the industrial revolution. Happiness, joy, pleasure, and satisfaction are fundamental aspects of human experience. We uniquely try to rationalize and explain joy with theories, hierarchies of needs and philosophical frameworks. Overanalyzing irrational objects and thinking ourselves into corners. Feelings, however, are often opaque and subtle. Overanalyzing emotion through perspectives and theories is akin to dissecting a frog. You understand it better, but the frog dies. Is there a need for this complexity: Is the secret to happiness in front of us?

Perhaps there is a simple truth: Enjoying what you have. Medvec, Madey & Gilovich's seminal paper argued that for Olympians achieving a bronze medal is more satisfying than a silver.[i] Their core argument is that the source of satisfaction from bronze medalists is from downward comparisons (i.e., thinking about a worse outcome) whereas silver medalists take on upward comparisons (i.e., thinking about a better outcome). In other words, counterintuitively, coming in a lower rank is more desirable from a satisfaction and happiness perspective. The bronze medalist is happy with what they have, the silver wants gold. This seems almost too simple; bronze is equivalent to satisfaction and joy.

While the authors craft this association, they also hypothesize a validation of this finding. Athletes who expect to continue to compete have an upward perspective. They focus on future performance. Per the hypothesis, silver medalists are likely to compete again. Bronze medalists are not. Using a dataset regarding Olympic athletes for the year they competed, and basic biographic data. We will investigate whether silver medalists are truly less happy than bronze medalists or indeed if a specific type of medalist is happier than others. Perhaps we will develop a recipe for happiness and success.

Data Description

The original Olympic dataset contains 15 variables and does not contain the target variable (future appearance) which was engineered using R scripting. While the data set contains a large amount of data on every athlete since 1896, there are some variables which are not relevant to this analysis. These are therefore removed from the dataset. Name, Team, City, and Games represent the athlete's full name, team, city of competition and a game played (e.g. 1992 summer) respectively. These objective descriptions help identify the athletes and their competitions, however, the information proved redundant. Specifically, ID is a more unique identifier than Name, NOC has fewer outliers than Team which contains many outliers such as China-1, China-2 and China-3, City is redundant as each Olympic game is played in a new and unique city, and finally, the data set splits out Season and Year making Games redundant. Therefore, I removed these from the model.

Based on the investigation I kept several fields and rows that may not be directly useful for the predictions. The Medal field contained many NA rows, for this analysis, I will assume that these athletes did not receive a medal and are intentional blanks. The descriptors of Sex, Age , Height , Weight, NOC, may prove interesting comparisons or covariant with the central question: do silver medalists re-compete at a higher rate than bronze medalists? The key fields for model and predictor engineering are ID, Year and Medal. Medvec et al. touch on a key idea in their paper. Some sports have bracket-style competitions where bronze and silver medals are earned as a direct result of winning against an opponent who comes in 4th or losing against an opponent who comes in

1st. The loss-win dynamic can conflate the feelings of satisfaction in the immediate aftermath of the competition. Like them, I will keep the Sport and Event fields to attempt to control this dynamic.

ID is a unique identification number for each athlete. The values seem assigned in an arbitrary order and vary from 1 to 135,571 indicating the total number of athletes in the data set. The athletes competed in anywhere from 1 to 24 events throughout their Olympic careers. 3 athletes were considered outliers based on the number of events they competed in: Aamodt, Alexandersson and Andrianov. Their primary sports were gymnastics and alpine skiing where there are many events that require similar skills. These outliers were kept. Year varied from 1896 to 2016. Olympic event attendance and participation have grown from 1896 to the 1990s when they plateaued in both the summer and winter Olympics. For the purposes of this investigation, all Olympic games are considered relevant. Few other outliers based on age in the data were noted but kept.

The average age of first-time Olympians is 24.4 with a standard deviation of 5.7. Age is normally distributed, as a population or when broken down by gender. (Appendix A) Similar trends hold true for the Weight and Height variables. Most Olympic athletes come from a few countries, the US, China, and European countries were over-represented. The asymmetric representation of the country (NOC) makes sense when considering population demographics and relative wealth. Olympic events require a complex logistical capacity and funding that wealthier, and larger countries can afford.

Based on the available data I created several fields to aid with the analysis. First, num_events_same_yr represents the number of events that the athlete has competed in within the same Olympic games. This field is useful to control for athletes that compete in multiple events and win multiple medals. Parsing the effects of multiple medals will obscure the relationship between satisfaction and medal. In a similar mindset, Medal_win_num captures the medal number of an athlete. Comp_num is the competition number of the athlete while prev_olypics is the number of previous Olympics. These may help understand past performance and relative standards for athletes. Similarly, New_sport and New_event capture if an athlete competes in several sports or events. This will help to differentiate athletes who compete in many sports or many events. Lastly, the Will_Compete_again field indicates if an athlete will compete in a subsequent Olympic event based on the data.

The data set will be split into 3 versions to test various parameters and theories. The first recipe is, the Original dataset with all predictors, to understand what influences athletes to compete multiple times. Next, the Spicy set only contains athletes who will win medals during their Olympic careers. Presumably, these athletes are the most competitive and may be different. Lastly, the Specialists set will only contain a subset of athletes who only compete in one event throughout their Olympic careers. This is because athletes are likely to compete in many events, earning several medals within the same year, which can be difficult to extract a clear meaning from.

A correlation plot indicated linear correlations between two groups. (Appendix B) The first group is Height and Weight which makes sense considering athletes are in peak physical condition and therefore likely have the ideal weight for their height and sport (outliers like weightlifting and wrestling exist). The second group correlated the maximum number of events with the number of Olympics in the career and the competition number, which may indicate that as athletes get older, they attempt new events.

For a more complete understanding, and to test predictions I also included 3 test athletes as separate datasets. Olympian 104659 will serve as the test. He was arbitrarily chosen. This Olympian competed once in 2004 and earned a silver medal. The only difference between the test sets is the medals earned. For test #1 they earned gold #2 silver #3 bronze. The test athletes will aid in the understanding of the model insights.

Model Selection & Methodology

To know the secret sauce of what makes an athlete satisfied we need sufficient depth of understanding. No one thing makes a person happy. With the same logic, no one model will sufficiently explain the sources of joy. We will use a combination of models to determine factors and indicators of what makes an Olympian re-compete. We will start with tree models (decision tree, random forest and boosting) backed by linear discriminant analysis (LDA), primary component analysis (PCA) and clustering analysis. Each of these models has strengths that will help in examining various aspects of the central question.

First, I will use 3 tree models to decipher the conditions that lead athletes to attend future games. We will use these insights to predict the 2012 Olympians who will compete in the future. In general tree models form decision trees, or a series of yes or no questions that help determine our predicted variable. Tress split the independent variables, (e.g. number of years since bronze, silver ect), based on how this impacts the dependent variable (future Olympics). The trees split if the accuracy of prediction is improved. When the tree splits based on improved accuracy, a node is created, and other sub-nodes are investigated. This process continues until the splits do increase the accuracy. We can control the splitting sensitivity and model accuracy through a variable called the complexity parameter. A high cp can lead to underfitting, a low cp can create overfitting. In both cases, the data is misrepresented by the decision tree, so we will aim to avoid an inappropriate cp.

The trees should demonstrate if receiving a silver medal is a strong predictor of repeat competition by having a decision node that indicates a silver medal as leading to “True”. To gain a more complete understanding of the dataset and what makes people complete multiple times, the model will also run random forests (a method of agglomerating many trees to help reduce bias) and boosting (a method that relies on the ‘wisdom of the crowd’ hypothesis). The nodes from the tree methods will help determine the most critical factors for re-competition rates, including medals won. Random forests and boosting methods generally have higher accuracy but are more difficult to interpret. Each of the three models will look at 3 datasets mentioned above: Original, Spicy and Specialists.

Next, I will use a technique called linear discriminant analysis (LDA) to predict re-competition based on the medal won. LDA uses statistical methods to make predictions based on prior probabilities. In other words, LDA investigates the probability of gold and repeat, using statistics and the probability of gold. This technique is perfect for this application as it will answer the question - knowing the medal won is silver what is the portability of re-competing? (Appendix C) LDA applied to the 3 datasets will predict if athletes compete again knowing which medal (if any) they have won. A variant of LDA is quadratic discriminant analysis (QDA). This method applies the same logic but with additional complexity as it assumes, a quadratic relationship between variables. We will apply LDA and QDA to both datasets to examine the medal to re-competition prediction.

As a third model, I will apply a PCA analysis. This technique is an unsupervised machine-learning model. In other words the model is not determining a prediction or desired dependent variable. Instead, PCA investigates the correlation or predictive power of variables against each other. The model assesses how much each variable contributes to a mean clustering center. In other words, PCA helps determine which variables contribute the most to differentiate each observation from the other. PCA can be difficult to interpret but several tools help make sense of the data. Specifically, we will rely on a visual and numerical analysis of the primary 2 components and which variables vary together. Understanding what makes an athlete want to participate in future games is an important factor in the dataset. PCA is a useful tool for knowing which variables are redundant or which ones will help determine the re-competition factor. We will also investigate the relationship between the number of medals

of each type received by athletes and the re-competition rate via PCA. If an athlete receives many silver medals throughout their athletic career, they may be more likely to re-compete against those who have received bronze. PCA will reveal data trends in this way.

The last model I will investigate is clustering analysis. This technique groups data points together based on their most important features. The clusters are points with similar characteristics. This method may allow for the identification of similar characteristics for bronze, silver and gold medal winners. Specifically, whether or not they are repeat Olympians, or what the characteristics of these athletes are if any.

Many of the above techniques allow data scientists to group data or make predictions based on the information and desired predictors. We will discover the associations between variables and use that knowledge to best understand what makes athletes compete repeatedly or retire happy. The predictive power of several tree algorithms will help predict the future appearances of 2012 athletes. The prediction will demonstrate the potential limitations based on the data available and complexity of the question.

Results

First, the tree-based models provided insights into the factors that help determine which athletes are most likely to compete repeatedly. Athletes who only had one event and were younger than 26 were the most likely to compete repeatedly. The models did indicate that Sport and NOC did have an influence on the re-competition rate. This is to be expected considering the wealth and support that some countries can offer their athletes and the physical intensity of some sports. (Appendix D) Among the subset of Athletes who have earned a medal during their Olympic careers (Spicy data), I examined the age, sex and medal won to predict their repetition rate. Age and sex were the top predictors indicated by the random forest. Women athletes under the age of 24 who won bronze were unlikely to compete in future events, while women athletes who won silver were slightly more likely to recompete, meanwhile, male athletes under 17 who had won bronze were very likely to recompete. This is a strong contrast to female athletes and to older male athletes between 28 and 41 who had won bronze or silver and were overwhelmingly unlikely to compete in future events. (Appendix E) The specialist and original data sets did not offer additional insights.

Random Forests and Boosting are more advanced tree techniques well suited for predictions. They both help understand the predictive power of each variable. Both models rely on different approaches for classification. However, they both reduce bias over tree methods. Indicators present in both serve as robust insights to use for predictions and feature importance. Both models indicate that for all athletes the main predictors of whether they compete repeatedly are not medals but rather age, height, weight, sport, and NOC. (Appendix F) A lesser consideration is the number of years since winning the first gold. Surprisingly medal ranks very low on the consideration of both random forest and boosting methods. This trend holds true for all the datasets. Although, for the champions in the spicy set the number of medals won had a stronger influence on their expected repeats than for the other sets. The test athletes illustrate the real difference between the medals for the likelihood of recompeting. Across all models, the boosted forest indicates the probability of future competition is highest when they earn gold, then bronze, then silver. Among the medals, winning bronze it seems is a stronger predictor of future appearances. Perhaps Bronze winners are not content after all.

LDA offered a different perspective on the data. Instead of looking at the factors and decisions based on nodes that split out, we can instead focus exclusively on prior probabilities. Or rather the repeat probability given the medal won. LDA determined that for all data set, athletes that earned a bronze medal have a 34% chance of not competing again and a 32.5% chance of coming back. Meanwhile, Silver medalists had a 33%

chance of not competing and 32.6% of competing. Therefore, gold medalists have a 33% chance of not competing again and a 36% chance of coming back. The linear coefficients of bronze and silver of -2.3 and -1.7 back up this trend as they indicated that silver and bronze medals have a negative association with the likelihood of future performances. The Quadratic version of LDA resulted in the same distribution. Bronze medalists are much less likely to compete again, and gold medalists are the most likely to return. These results indicate that all else equal bronze medalists retire often. This trend is reinforced when comparing medals won by Olympians against the number of Olympics competed in. As athletes compete in more competitions, regardless of the number of previous Olympics, they are more likely to win Silver. (Appendix G) LDA helps to better understand the likelihood of competing given the medal won, which is our focus. LDA reviled one additional insight when comparing Height and Weight against repeated Olympics, athletes who are heavier than the population of Olympians are significantly more likely to repeat their performances.

PCA relies on grouping data based on the predictive power of the variables. Based on the original data set Year, Height and Weight were all highly correlated with each other. Likely indicating better sources of nutrition and food security since 1896. Year and the number of the medal won are negatively correlated, indicating that as time has moved on the Olympics have generally become more competitive, similarly athletes compete in few events the heavier and taller they are. This may be due to sports with many events like gymnastics or ice skating, have many events, as opposed to strength-based competitions like shotput or weightlifting which have fewer competitions. (Appendix H) When we investigate Olympic champions, there are a few differentiators in the grouping. For the subset who have previously won a medal, we introduced an indicator field to represent the certainty that they will compete again or not. Height, Weight and future competition fields were highly correlated. This is likely due to the previous findings whereby gold medalists tend to re-compete more and be physical outliers. Age and the number of events within the same year were negative indicators of re-competing. As athletes get older and compete in more events they compete less. Surprisingly the number of medals that the athletes won had no bearing on any variables. When comparing the number of each type of medal won and the future competition indicator it was clear that silver and gold wins correlate with the likelihood of recompeting. (Appendix I) The specialist data set demonstrated this most clearly. PCA would seem to indicate that many factors are colinear and important. The medal earned has a small role to play in the likelihood of athletes competing in the future.

Lastly, a clustering algorithm looked to understand the inter-operability of the variables for the datasets. There is a possibility as suggested by the tree models that different ages have different characteristics and priorities. The clustering mode will reveal this. I used a clustering of 8 to give sufficient differentiation between clusters. Based on the central question I plotted years since the first medal of any type against the number of the Olympics competed in at that stage in the athletes' careers. The clusters revealed that many repeat Olympians had won in their first year. Across the range of the first medal won (bronze, silver and gold) the athletes had similar characters for the 2nd or 3rd Olympics in their careers. However, as the number of Olympics in athletes' careers increased, the clusters differentiated. The athletes who won a medal in their first year are seen on the diagonal in (Appendix J). Athletes who won Bronze Medals in their first Olympics tended to belong to cluster 8 at first and then cluster 6 later in their careers. Cluster 6 is a group of athletes who have fewer medals throughout their careers (1.3 medals vs. 1.6 for other clusters) and competed in 3 Olympics on average. Athletes who won silver medals in their first year tended to belong exclusively to cluster 8. This cluster won more medals throughout their career (1.8 medals) and was much less likely than other groups to have won bronze. Lastly, athletes who won gold in their first year were split between cluster 8 and cluster 5 throughout their careers. Cluster 5 won the average number of medals (1.6) and had a higher number of athletes who won bronze in their first year compared to other clusters.

These methods have given different insights into the dataset and potential implications for athletes, their happiness and expected future performance.

Predictions & Conclusion

Explaining or predicting emotions perplexes us and is almost beyond our grasp. The Olympic dataset and the models explored have generated complex or even conflicting ideas on what medal brings happiness. However, there are unique trends. First, for the athletes that won medals of any kind, the most likely to return to the Olympics were the gold medalists, not silver or bronze. Potentially, implying that the gold medalists feel the most pressured to repeat their performance. Second, there are much better predictors to determine if an athlete will re-compete in an upcoming Olympics than their medal won. Lastly, the models confirm the suspicion that silver medals tend to influence athletes to compete in the future more than bronze medalists. The significance of this is rather low.

The LDA analysis revealed that gold medalists are the most likely to compete in future events compared to silver and bronze medalists. The boosting tree model reinforced this as a sample test of athletes 1,2,3 demonstrated that all else equal, gold medalists were the most likely to compete again. The most likely explanation is that athletes who compete and successfully win gold are pressured into repeat performances. The pressure may be internal or external from their host country. For instance, Simone Bile, a gold medal gymnast dropped out of the Tokyo games as it was revealed that she was coerced into competing for the US team based on her 2016 success. [ii] On a similar thought, the boosting model compared the influence on repeat performances and found that amount the number of years since the first medal predictor, years since the first gold was in fact the most influential. An athlete who won a gold medal in several games in the past is a stronger predictor than any other medal. Examining the KNN model reveals a similar trend: athletes who won gold their first year are the most likely to have attended many Olympics throughout their careers.

The second insight that the models elucidated is that there is a complex series of factors and decisions that influence the number of Olympics that an athlete will compete at. In general, the primary factors were age, NOC, sports, height and the number of events. The boosted tree model was trained on the pre-2012 data to predict the 2012 repeated athletes. The mean square error of the boosted model with most variables was 0.32 whereas the mse of a boosted model solely based on medal and age mse rises to 0.37. For context random guessing would have an mse of around 0.5. This demonstrates that the repeat participation rate in the Olympics is subject to so many variables and uncertainties that examining the medal awarded is a particularly difficult challenge that is overshadowed by other factors. While this may taint the purity of the competition, there are certain realities that justify this complexity. Some athletes are injured because of the intensity of their sport, while others have difficulty growing and developing. There are 4 years between the Olympics and outside factors like family or financials often play a huge role in re-competing. The Olympians who win advertising deals or do not have to struggle financially are the exception to the rule.

The final insight revealed that the researchers were correct, at least to some degree. It depends on the perspective and the model. In general, silver medalists have higher odds of competing in future games than bronze medalists, but it depends. This is the least clear insight from the models. The tree models indicated that other factors like age or gender are far more important in the consideration of repeated performances. When factoring in those considerations young athletes are likely to appear in future Olympics. For this group earning a bronze medal indicates repeated performances more than silver or gold. However, for athletes above the age of 24, athletes are more likely to compete if they win silver than if they win bronze. The LDA analysis demonstrated that all else equal silver medalists are slightly more likely (32.5% Vs. 32.6%) to re-compete than bronze medalists. The small difference does not control age which could help explain this deviation. The age factor is likely related to brain growth and development, and parental pressure. While athletes are young, they

are more easily influenced and can be brainwashed into believing that gold is the objective for optimal satisfaction. The authors of the study controlled this by investigating the desired performance of athletes before competing. We cannot control this. The PCA does, however, give insights into which factors best cluster the data. When examining the campion athletes who have won at least once, PCA demonstrates that winning silver or gold is most closely correlated with repeated performances. The association is relatively small but does demonstrate this link. The association between repeated performance and medaling is not a clear one. While data science methods demonstrate a very small relationship between the medal earned and repeat performances it may not yet be able to incorporate all aspects of the Olympic decision-making process.

4 models with 3 key insights. Data science is not a magic cookbook for success where you follow a recipe and add data as the ingredient. There is no clear recipe for happiness. Using advanced computational methods to investigate long-held beliefs on who is the happiest after an Olympic game gave different perspectives and insights. Ultimately there is no one factor that can determine success or joy. The human condition is a complex one with many facets and dimensions. Perhaps Manson was correct when he proposed that happiness comes from accomplishing your personal agenda.[iii] To many, the games are a display of athleticism and bravery. For the athletes, they can be the ultimate source of joy or pain. Winning gold is a double-edged sword as it seems to bring the ultimate joy, but with the pressure to perform again. To me this reflects the human spirit to always seek more but also a lesson in life: enjoy the moment, enjoy what you have, enjoy the little things.

Appendix A

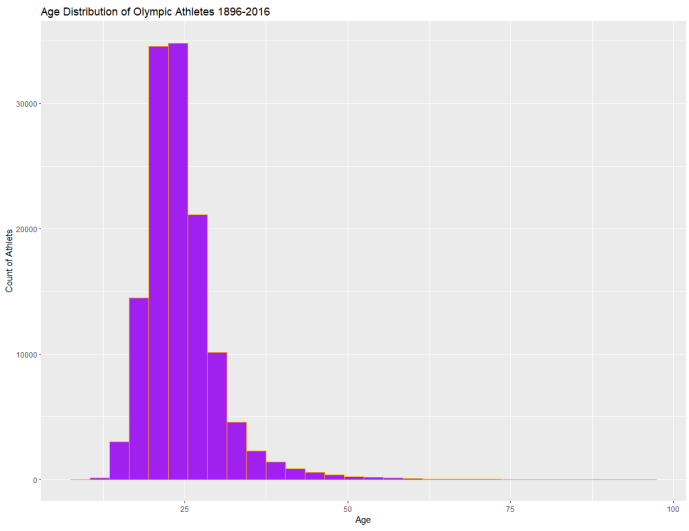


Figure 1: Distribution of Olympic Athletes' ages

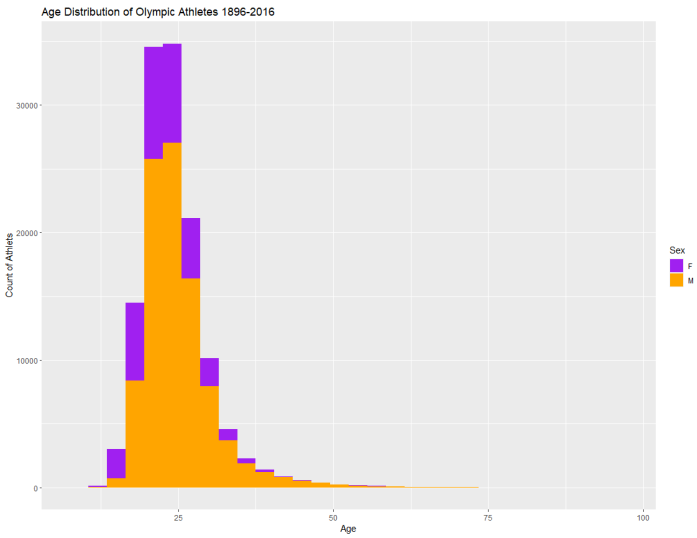


Figure 2: Distribution of Olympic Athletes' ages

Appendix B

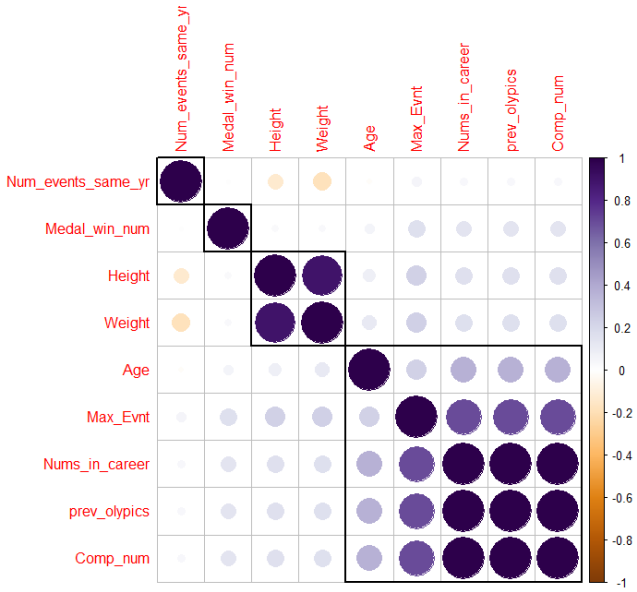


Figure 3: Coreation Matrix for numerical variables. Note: A correlation of 1 or -1 indicates a perfect linear relationship. 0 indicates no relationship

Appendix C

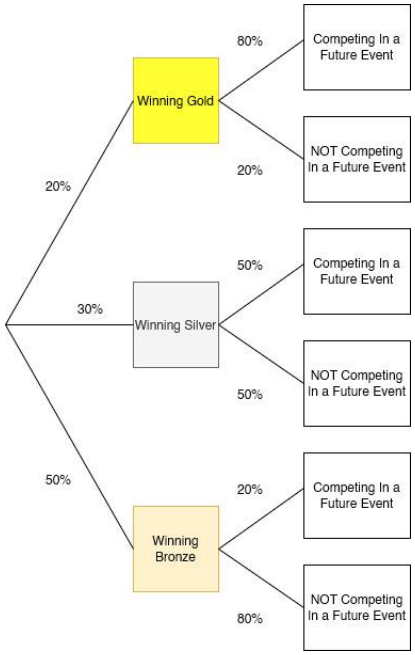


Figure 4: Example of prior probabilities and statistical framework behind LDA process

Appendix D

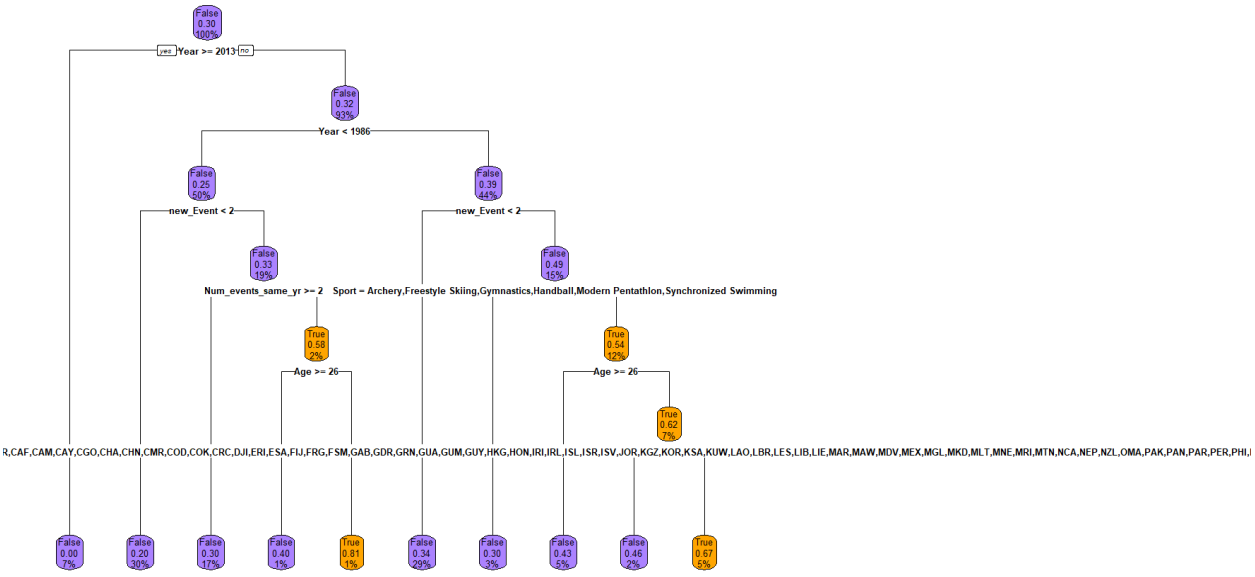


Figure 5: Decision Tree of Original Data Examining Will_Compete_Again (True/False). Note: complexity parameter=0.004

Appendix E

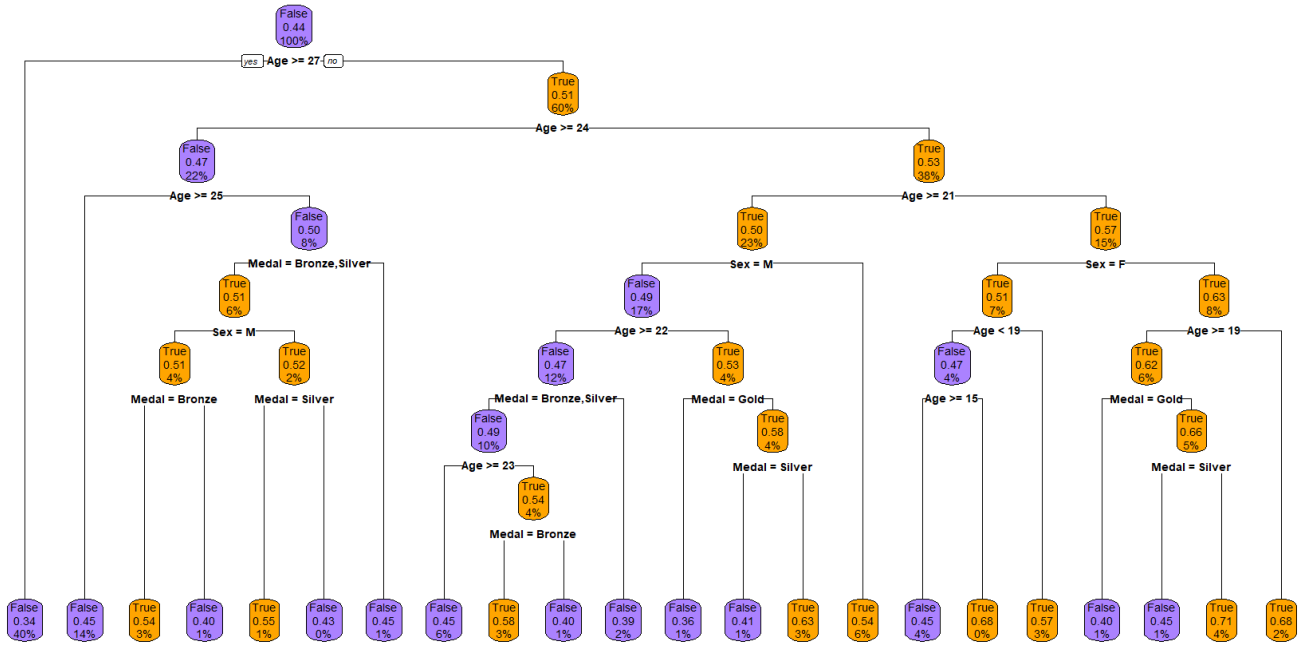


Figure 6:Decision Tree of Spicy (Athletes who have a least 1 win in their careers) Data Examining Will_Compete_Again (True/False). Note: complexity parameter=0.001

Appendix F

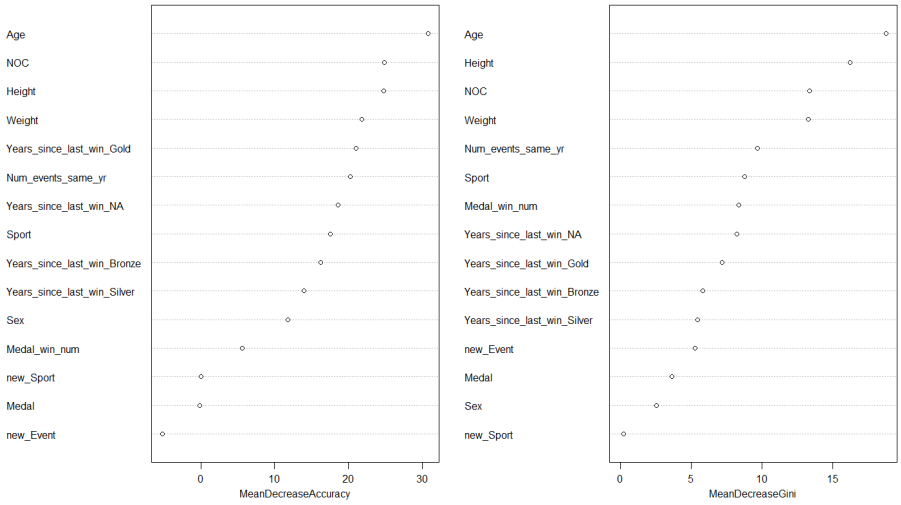


Figure 7:Importance plot of Random Forest on Original Data Examining Will_Compete_Again (True/False). Note: Higher mean decreased accuracy indicates more variable importance. Higher mean decreased Gini coefficient indicates less homogeneity and more model import

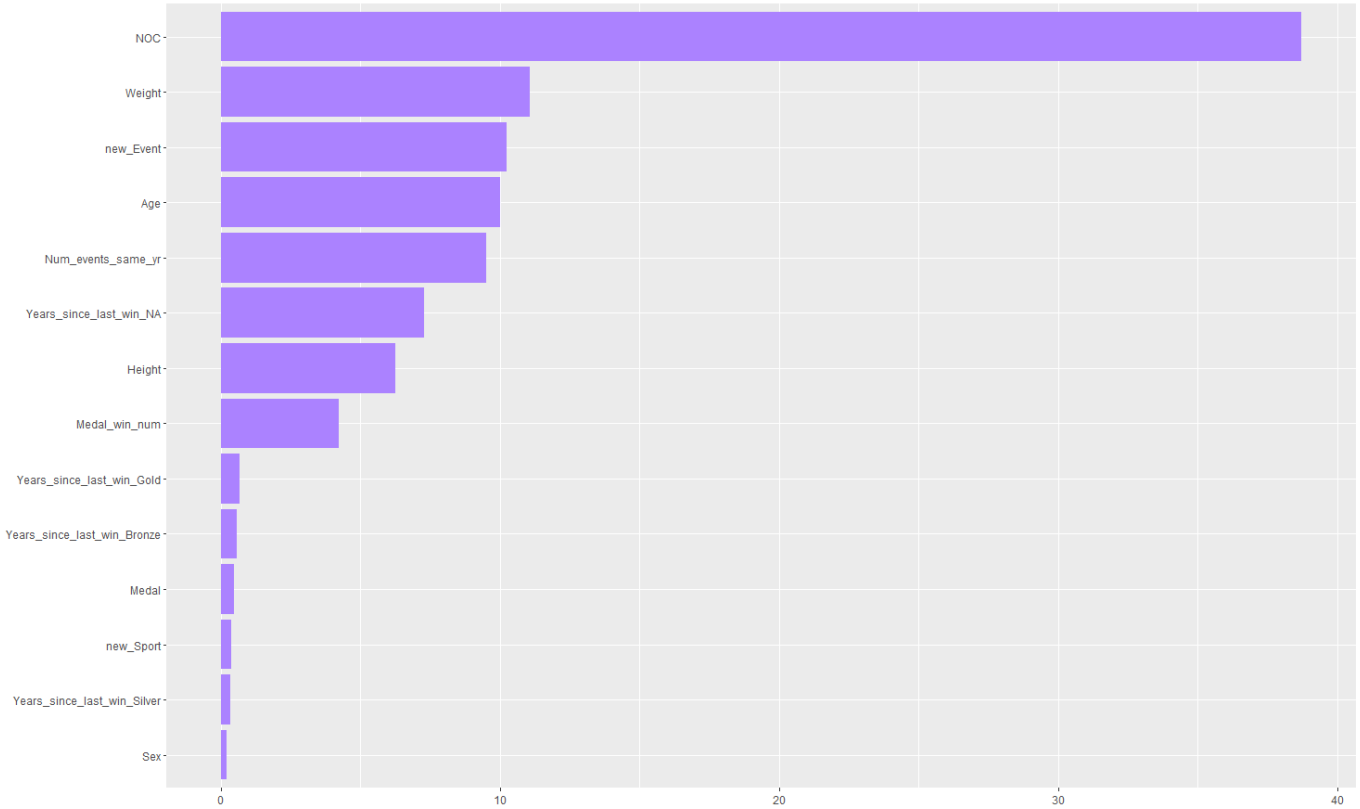


Figure 8:Feature importance of Boosted Forest Examining Will_Compete_Again (True/False) Note: Scale is relative importance

Appendix G

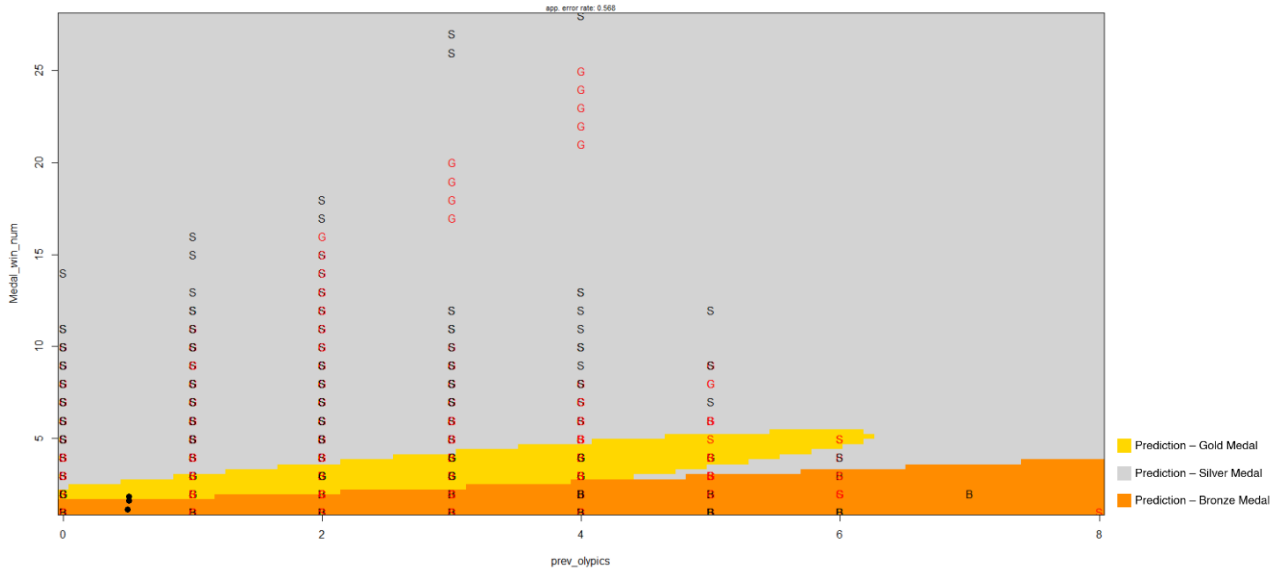


Figure 10: LDA Partition Plot of previous number of Olympics vs. Number of medals won. Note: Colored regions correspond to predictions of Medal types. Letters in black represent correctly classified observations, red incorrect ones.

Appendix H

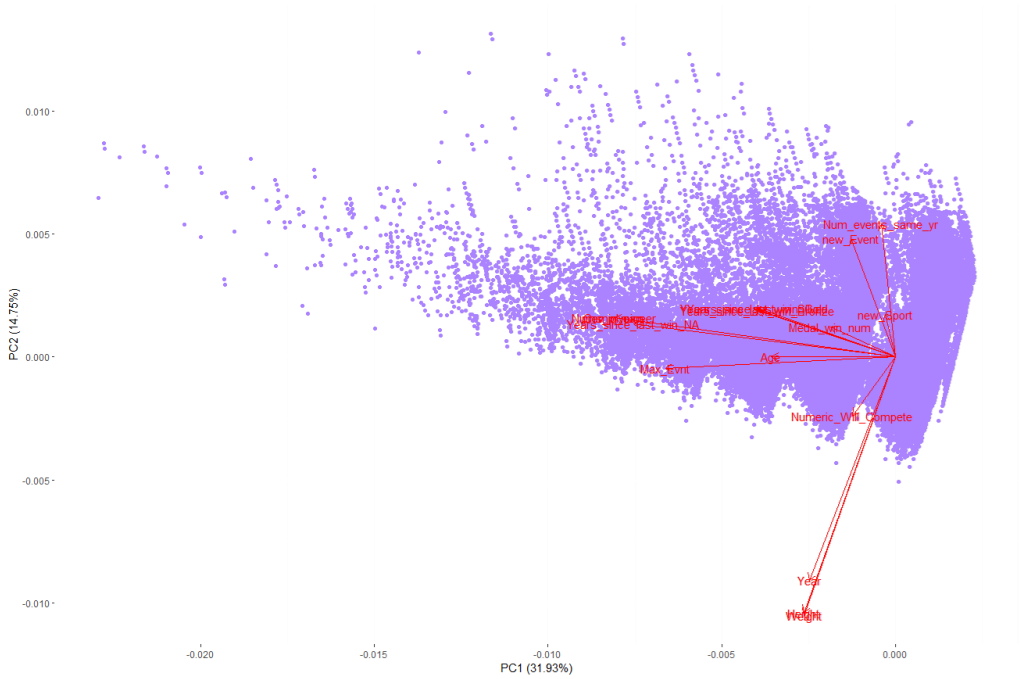


Figure 11: PCA plot of all factors in Original Dataset. Note: red arrows represent the importance of each predictor. Red arrows that overlap indicate collinearity arrows that are in the same direction indicate correlation

Appendix I

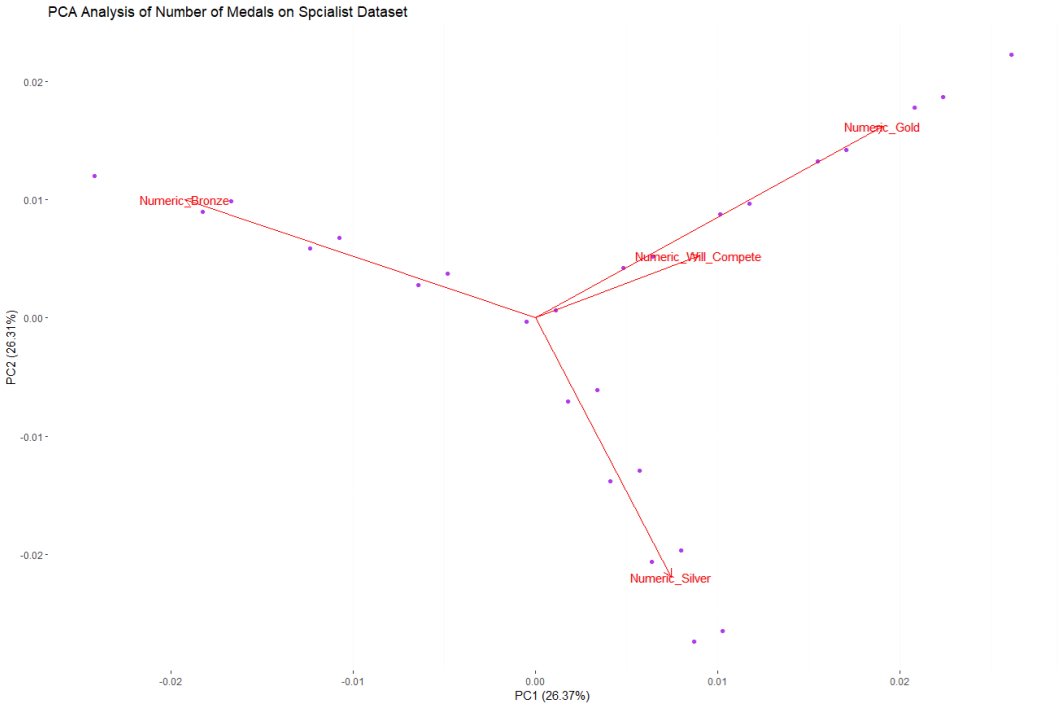


Figure 12: PCA plot of recompeting indicator (1/0 for True/False), and total number of type of medal won in Specialist Dataset..

Appendix J

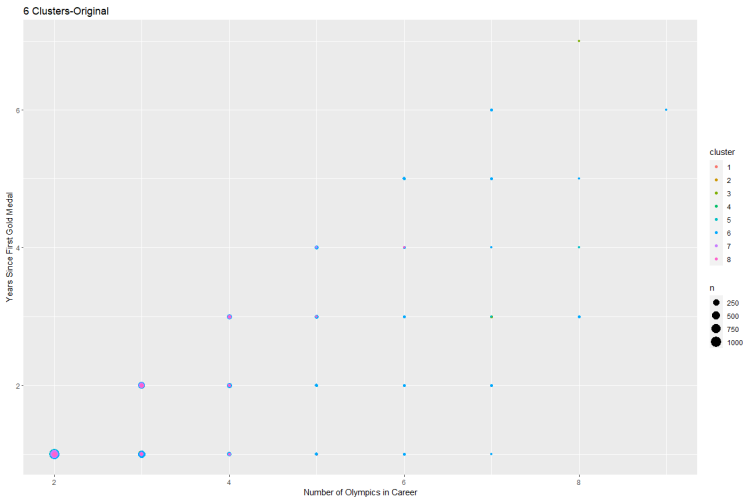


Figure 13: Clustering analysis of years since bronze win vs. number of Olympics in career

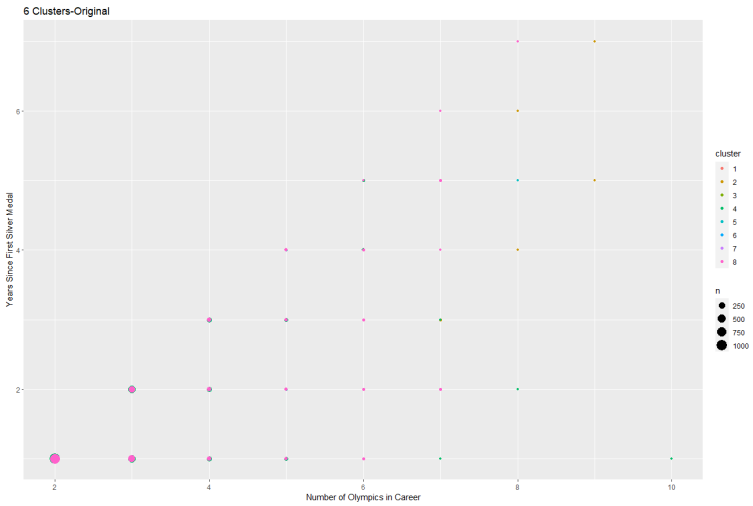


Figure 14: Clustering analysis of years since first silver win vs. number of Olympics in career

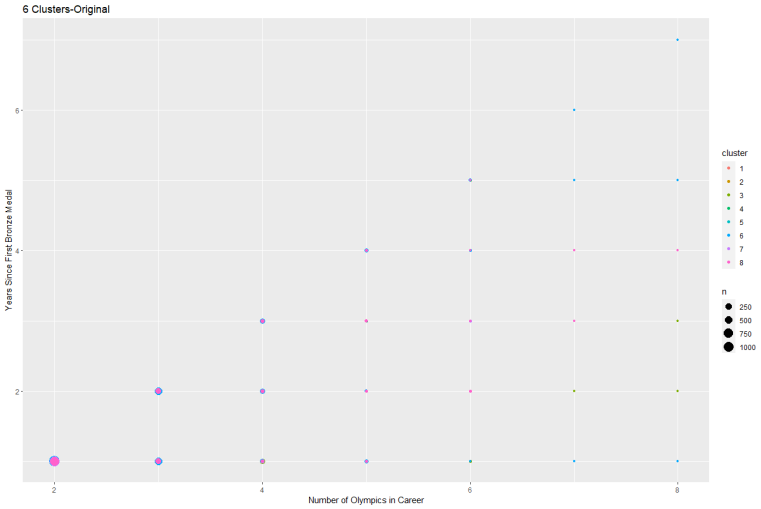


Figure 15 Clustering analysis of years since gold win vs. number of Olympics in career

References

[i] Medvec, V. H.; Madey, S. F.; Gilovich, T. When less is more: counterfactual thinking and satisfaction among Olympic medalists. *Journal of personality and social psychology* **1995**, *69* (4), 603.
[ii] Chappell, B. Simone Biles Reminds Us That Even Superstars Feel The Pressure. *NPR* **2021**.
[iii] Manson, M. *The Subtle Art of Not Giving a F* ck: A Counterintuitive Approach to Living a Good Life*; Macmillan Publishers Aus., 2016.